Revision of Basic Probability Transcript

Hello and welcome everyone. Today we are going to explore the fundamentals of probability. This is the mathematics of chance and understanding it allows us to quantify uncertainty and make better predictions.

This is a crucial skill in games, science, finance, and also they are used in LLMs that we use every day, right? Today, at least today. So let's understand and try to quantify uncertainty using the basics of probability. What exactly is probability? Probability at its core is a way to measure how likely an event is about to happen or how likely an event is to happen.

We express this with a number on a scale from zero to one. Zero meaning it is absolutely an impossible event. One means it is a certain event and 50-50, which is 0.5, is a chance, right? Either of them can happen.

Either it could be zero or it could be one, somewhere in between, right? So chance is in between. Now probability is defined as total number of favorable outcomes divided by total number of outcomes. So what could happen is at the bottom and favorable is at the top.

Numerator is favorable and all the possibilities that could happen is at the bottom or it is at the denominator. Now let's understand this using a simple example. Now if you see here, we have a fair coin.

A fair coin is in which you have equal chances of getting a head or tail, okay? Now you see a coin here which is currently faced at a tail. A head is when you see someone's face on it, right? Now the probability of getting a head or tail in a fair coin is 0.5. Now let's say if I flip the coin once and I am trying to observe the probability of head. Right now it is represented as n slash a but once I flip it once, okay, see there is the first flip itself is head, okay? Now let's say if I flip it again, I see that there are three flips, two tails, one head and the probability is 0.33 because out of three flips there is only one head, right? Now once I flip it again, you see that there are four flips altogether but the number of heads is one because again you have got a tail.

Now as you repeat this experiment multiple number of times, many number of times, let's say if I repeat it another hundred number of times, now you have total flips as 104, total heads as 59, total tails as 45. Now you see that as I keep pressing this flip 100 coins button again and again, you would see that probability of heads or observed probability of heads would get closer and closer to 0.5. This is called as law of large numbers. This is something you will be studying in estimation and sampling theory later on but at least you are able to guess get an idea about how making things occur multiple number of times you and judging it and taking some analysis out of it or making some analysis on top of it, right? You get to a number which is something you are likely to see or you are aware about, right? A coin flip, you already know that there are 50-50 chances of getting a head or tail.

When you repeat that experiment multiple number of times, what you see is there are equal number of chances of getting a head. Right now you see observed number of probability of head is 0.51 which is almost 0.50. Now because we have studied about all of those things, let's define some key things within probability. The very first thing that you should understand is a random experiment.

A random experiment is something which has random outcomes, okay? So let's say when I flip a coin, the outcomes are but either of these are random. It can result in either head or tail randomly. That's why it is called as a random event.

The outcomes of or all possible outcomes of random experiment is called as a sample space. Now when you roll a dice once, these are all possibilities that you get. 1, 2, 3, 4, 5 or 6, right? You cannot get lesser than 1 or more than 6 because a 6-sided dice is only having these numbers on its face, right? So sample space is all possible outcomes of a random experiment.

While an event is a subset of some sample space. Let's say if I say what is the probability of getting an odd number? Now there are these 1, 3 and 5. These three are odd numbers. The probability of getting an odd number would be 3 divided by total number of outcomes or what could happen which is 6. 3 by 6, it would be 0.5. Similarly, here we have demonstrated the probability of getting an even number.

Even numbers are 2, 4 and 6. Hence the probability of even is equals to favorable which are 2, 4, 6 divided by all possible which are all of these sample space. 3 by 6 is 0.5, right? Now this vane diagram helps us visualize a lot of probabilities. Let's say if we are interested in defining what is the probability of two events occurring, right? A union B or A and B. A intersection is what is the common probabilities or what is the joint probability between them.

A union B is either event. Let's say what is the probability that I will go out and play and also eat ice cream. That is A intersection B. When I say what is the probability of I going out and play and also or also not going out for studies.

That is A union B because the term or or either is represented in it, right? Or or either represented represents union while when I say and it represents intersection and that's what is visualized here. If you see A and B as both events and you see that this intersection term in between that's a common part which is called as intersection. Now what exactly is conditional probability? Conditional probability is probability of occurrence of an event given another event has already occurred.

So let's say probability of rain given it is dark clouds in the sky. So that is conditional. Now you already know that there are dark clouds.

What is the probability of having a rain, right? So that exactly is what we define as conditional probability. Probability of having a rain maybe that could be represented as A given B which is the dark clouds. Now you have already seen dark clouds.

How does the probability of rain changes with the new evidence that you have got is what is called as conditional probability. Now probability of A given B is represented as how many times or total number of times both have happened together A intersection B divided by probability of B okay which is in our example probability of dark clouds and rain happening together in the data or in our historical data whatever data that we have or whatever evidences that we have till now divided by how many times dark clouds have happened right B is dark clouds that's what it represents. So conditional probability is probability of an event given another event is already has happened or already has occurred.

Now this is an interactive demo just to help you understand conditional probability. Let's say I have a very small deck of eight cards. Now I am not representing everything 52 faces on the screen because it would be very tidy or not very much clear on the screen to visualize it right.

Let's say that I have these eight different cards or deck of cards and let's find out the probability of drawing a king given the card is a face card right. Now how many face cards are there on any deck there is either a king or a queen or a jack there are these three face cards right if you have ever played cards at your home or you already know someone who plays it so there could be either king queen or jack these are the face cards. Now because I have a small deck of eight cards which are represented here if you see if I could click show full deck you see that this is the entire full deck.

Now if I apply condition that the card has to be a face card these are the only face cards king king of hearts king of spade queen of heart right so all of all of them has been represented here king queen joker queen king right now you have five possibilities there are five face cards out of these eight card deck.

Now let's apply probability of king given face card now let's say out of these five how many are king there are only two right so the entire probability would be two divided by the number of face cards that you see that's what you see these are two kings and the probability is two by five which is 0.4 right so in totality conditional probability is when you already know about something what is how much by how far the another probability changes so likelihood of an event a given another event b has already occurred so I hope this was able to help you understand the basics of probability now we'll dig a little more deeper into statistics and sampling distribution so on but at least

I should ensure that or I should this entire video should ensure that you have understood the concepts of basics of probability. Thank you everyone.

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